Isaac Denning

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Education

Iowa State University – B.S. in Software Engineering, GPA: 3.81 Expected Graduation: May 2026

Work Experience

Kingland Systems: *May 2025 – August 2025*

- Worked on an Agile team to fix and improve a full-stack application
- Added S3 bucket to the project's AWS infrastructure via Terraform configurations
- Updated system to use auto-generated OpenAPI code for API connections
- Updated checkstyle rules and researched tools to automatically fix checkstyle errors
- Fixed general defects in the project's Angular frontend and Spring Boot backend

Collins Aerospace: *May 2024 – December 2024*

- Updated the Boeing CH-47F Chinook Control Display Unit with a team of engineers
- Reviewed Boeing contracts, translating them into system requirements, then into tasks
- Used parallel development to write code and tests that meet system requirements
- Setup server to automatically parse test logs and publish results to web interface
- Automated our Control Display Unit load process for development

Herzog Technologies: *June 2023 – August 2023*

- Worked with an Agile team to develop two separate programs
- Created a C# program to monitor the location and safety of rail workers
  - Developed various geofencing algorithms
  - Built a connection to the GPS unit
- Created a full-stack application for contacting emergency personnel in case of a rail accident
  - Configured Angular frontend, .NET backend, and dockerized infrastructure
  - Created Azure pipeline for building, testing, and deploying the website

Ricker Lyman Robotics: *October 2020 – August 2022*

- Automated multiple OS load processes for the server units
- Created stress-testing scripts for the server units
- Developed a program to transfer data from sales databases to the inventory database
- Troubleshot and fixed general server issues

Activities

Cyclone Search and Rescue:

- Worked on a team to create a robot for a search and rescue competition
- Transformed codebase to implement Robot Operating System
- Built website for viewing motor positions and camera feed

Scouting America:

- Eagle Scout Project: Led a group to create a patio at Camp Io-Dis-E-Ca
- Ranger Project: Constructed a bee shed and introduced woodcutter bees to Allen's Orchard

Personal Projects:

- Designed and built a machine learning AI which scored 94.3% on the MNIST test
- Programmed a permutation search chess bot with a minimax decision rule
- Built a programmable 8-bit computer from basic logic gates on an FPGA board
- Contributed to the open source project Zulip, a Discord alternative
- Developed a testing framework for C with custom compiler attributes

Technical Skills

Languages:

- C, C++, C#, Java, Python, TypeScript/JavaScript, HTML/CSS, Ada, Bash

Programs:

- Git, Docker, Spring Boot, Angular, PostgreSQL/MySQL, Terraform, Subversion

Miscellaneous:

- Linux, Windows
- Technician class radio license KE0HED